Topic 6 – Interpretation of accounts – Solutions

Question 1 – Hedge

Dividend yield: Dividend per share X 100 Market price per share	 7 X 100 350	 2%
Dividend cover: Profit for the year Dividends for the year	 70,000 0.07 x 500,000	 2 times
Earnings per share: Profit for the year Number of shares	 70,000 500,000	 14 pence
Price/earnings ratio: Market price per share Earnings per share	 3.50 0.14	 25

Question 2 - HJK Ltd

Accounting ratios:

	2020	**2021**
Gross profit margin	525 x 100 = 35% 1500	600 x 100 = 31.6% 1900
Operating profit margin	275 x 100 = 18.3% 1500	250 x 100 = 13.2% 1900
Return on capital employed	275 x 100 = 27.5% 1000	250 x 100 = 20% 1250
Inventories turnover period	100 x 365 = 34 days 1075	200 x 365= 49 days 1500
Current ratio	500 = 6.3:1 80	1000 = 4.8:1 210
Trade receivables settlement period	375 x 365 = 92 days 1500	800 x 365 = 154 days 1900
Trade payables settlement period	80 x 365 = 30 days 995	200 x 365 = 53 days 1400

Comments:
The company increased its sales in 2021 by £400 million (26.7%). It seems to have done this by reducing its profit on goods sold, as the gross profit has reduced. The increased activity probably accounts for the additional expenses. As a result, its operating profit margin fell to 13.2%, and the operating profit was reduced by £25 million.

HJK plc's liquidity position is still healthy, as indicated by the current ratio, although it has fallen from 2020. The trade receivables settlement period has increased substantially (as has the trade payables settlement period). This may be as a result of a deliberate policy to stimulate sales, or it may reflect poor receivables management. Not surprisingly, the cash position has deteriorated over the period, with the company having an overdraft by the end of 2021.

Increased trading activity does not always ensure a company's survival. If the company cannot pay its debts as they fall due, it may be unable to continue to operate. HJK plc should pay close attention to its cash flow position.

Question 3 – X and Y

Profitability
ROCE is much higher for Business X than for Business Y, indicating a more efficient use of its assets. For every £1 invested in X, investors are obtaining a return of 20p, compared to only 12p in the case of Y. Perhaps less storage space is needed to display the products on sale by X, resulting in a smaller investment requirement in expensive non-current assets.

X has a much higher gross profit margin than Y. It may be assumed that Y concentrates on volume, while X has a higher mark-up. It may be that X serves a smaller clientele that is prepared to pay higher prices.

The operating profit margin is the same for both businesses, reflecting the fact that the more exclusive business will have relatively higher operating costs.

Liquidity
The current ratio and acid test ratio both show that X has a higher working capital requirement than Y. Comparing the current and acid test ratios, Y has higher inventories relative to liabilities than has X. It may be inferred from this that X has significantly higher trade receivables. This reflects the fact that a more up-market retailer is far more likely to give extended credit to its more exclusive, and richer, customers.

Efficiency
The trade receivables settlement period also indicates higher levels of trade receivables in X, who enjoy an average credit period that is three times than of Y.

The trade payables settlement period is higher for X than for Y. This could be a sign of a business having cash flow problems, but in this case it is more likely to indicate that X has managed to negotiate longer payment periods with its suppliers.

Inventory moves more slowly in X than in Y, perhaps reflecting the less mass-market nature of the products sold by X.

Conclusion
The lower margins and faster inventory turnover of Y would suggest a high sales volume at low prices, while X clearly sells more slowly at a higher mark-up, suggesting a niche rather than a mass market.

Question 4

Yes, provided the underlying information has been accurately and consistently prepared, that the basic deficiencies of such underlying information are understood, that the ratios themselves are prepared on a consistent basis, and that they are then used as a guide to what might have happened.

Ratios give some indication of what and where certain events may have occurred, but the clues that they provide need to be followed up. It is not sufficient to argue, e.g. that trade receivables settlement period is now 100 days compared with (say) 50 last year. This apparent fact provides some valuable information but it is then important to investigate and to explain why the trade receivables collection period has doubled in such a short time period.

So, it is important to use ratios to *interpret* what has happened and not to assume that a list of ratios is sufficient an explanation in itself.